Momentum analysis: put it together

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a diagram or calculation to compare a deflection option. Explain where your prediction stops working.

SUCCESS CRITERIA - I can...

1. Compute the required change before evaluating an option.
2. Keep impulse, velocity change and displacement distinct.
3. Give the numerical result and state what the calculation treats as true.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A system's total momentum is **15 kg·m/s** *west* and **21 kg·m/s** *south*. Its combined mass is **3 kg**. **Find final velocity components**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: shift = Δv × time; J = MΔv.
Fictional calculation that treats the motion as a straight line. M is target mass; J is required impulse. This is not a real encounter forecast.

Worked reasoning: A **1 kg** cart at **+6 m/s** sticks to a **2 kg** cart at rest. Final velocity is **+2 m/s**. The moving cart changes momentum by **-4** and the other by **+4 kg·m/s**.

Your turn: Draw both momentum changes and **show how their sum matches** the system change.

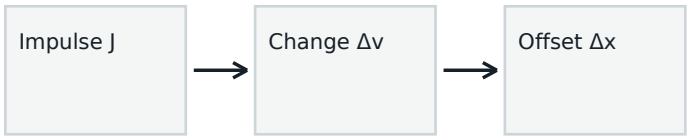
Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

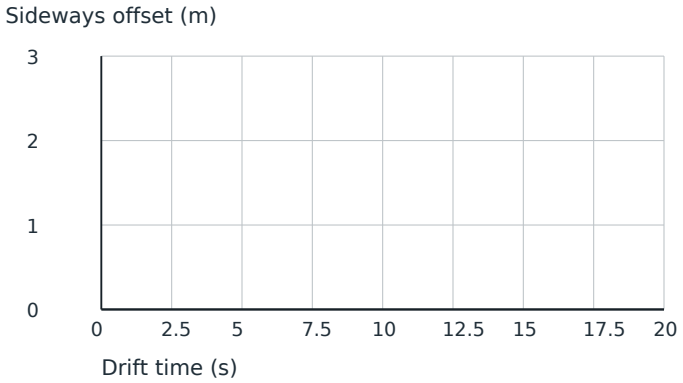
Organize this chain with units: impulse → velocity change → sideways shift. Which step uses target mass?



Marzano check - circle: ① ② ③

04 Apply the physics

In a simplified deflection simulation, a **100 kg** target needs a **2 m** sideways offset after a **20 s** drift. Assume the added sideways velocity stays constant after a brief push. **Find the required velocity change**, then the impulse.



Marzano check - circle: ① ② ③

Momentum analysis: put it together

antocciphysics

Continue down each column.

05 Explain your reasoning

If the target mass estimate is too low, how would a fixed delivered impulse’s velocity change compare with the prediction?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** cart moving at **6 m/s** stops in **3 s**. **Find momentum-change magnitude** and average stopping-force magnitude.
Use before and after snapshots to organize the momentum account.

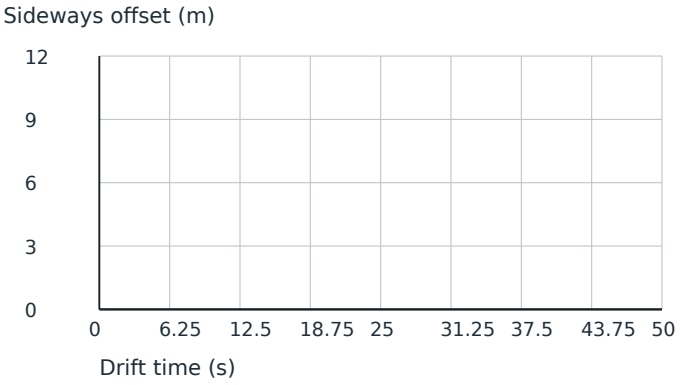
Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A brief push adds **0.2 m/s** sideways to a probe’s velocity. The probe then drifts for **50 s**. **Complete the position-time line**, find its sideways offset and **state one thing you** are treating as true.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. The same pair now receives an external impulse of **3 N·s** right during contact. **Find final total momentum** and shared velocity. Organize the known quantities on the before-and-after diagram.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3